



Dissertation on

“Leveraging VQA to Analyse Autism Spectrum Disorder (ASD)”

Submitted in partial fulfilment of the requirements for the award of degree of

**Bachelor of Technology
in
Computer Science & Engineering**

UE20CS390B – Capstone Project Phase - 2

Submitted by:

Charvi Bannur	PES1UG20CS638
Chaitra Bhat	PES1UG20CS635
Gagan Goutham	PES1UG20CS550
Sada Kakarla	PES1UG20CS586

Under the guidance of

Prof. Mamatha H R
Professor
PES University

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
FACULTY OF ENGINEERING
PES UNIVERSITY**

(Established under Karnataka Act No. 16 of 2013)
100ft Ring Road, Bengaluru – 560 085, Karnataka, India



PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)
100ft Ring Road, Bengaluru – 560 085, Karnataka, India

FACULTY OF ENGINEERING

CERTIFICATE

This is to certify that the dissertation entitled

‘Leveraging VQA to Analyse Autism Spectrum Disorder (ASD)’

is a bonafide work carried out by

**Charvi Bannur
Chaitra Bhat
Gagan Goutham
Sada Kakarla**

**PES1UG20CS638
PES1UG20CS635
PES1UG20CS550
PES1UG20CS586**

in partial fulfilment for the completion of seventh semester Capstone Project Phase - 2 (UE20CS390B) in the Program of Study - Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period June - December 2023. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 7th semester academic requirements in respect of project work.

Signature
Dr Mamatha H R
Professor

Signature
Dr. Mamatha H R
Chairperson

Signature
Dr. B K Keshavan
Dean of Faculty

External Viva

Name of the Examiners

Signature with Date

1. _____

2. _____

DECLARATION

We hereby declare that the Capstone Project Phase - 2 entitled “**Leveraging VQA to Analyse Autism Spectrum Disorder (ASD)**” has been carried out by us under the guidance of Prof. Mamatha H R, Professor and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology in Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester June - December 2023. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

PES1UG20CS638

Charvi Bannur

PES1UG20CS635

Chaitra Bhat

PES1UG20CS550

Gagan Goutham

PES1UG20CS586

Sada Kakarla

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ABSTRACT

Diagnosis of psychological disorders is an open-ended critical problem exacerbating the direct effects it poses to the mental health condition of a person which if not addressed would manifest itself silently affecting their thoughts, emotions and behaviours. Modern state-of-the-art models have been intrinsically sensitive to various parameters, which reflects the extent of human labour and confines itself to the identification of a single disease often trained using redundant data. The purpose of this project is to create a goal-oriented Visual Question Answering (VQA) system that indicates the degree of the Autism Spectrum Disorder (ASD) that could be used in the early stages of diagnosis. Using a unique VQA architecture, the system analyses the video of patients with a specific disorder, extracts characteristics, and correctly predicts the extent of, existence of the disorder. The VQA system has the potential to improve the accuracy of diagnoses, monitor treatment effectiveness, reduce the time and resources required for diagnosis, provide an objective and accurate diagnosis to aid in the treatment and also generate new insights.

TABLE OF CONTENTS

Chapter No.	Title	Page No.
1.	INTRODUCTION	1
2.	PROBLEM STATEMENT	4
3.	LITERATURE REVIEW	6
4.	DATA	14
	4.1 YOUTUBE VIDEOS	
	4.2 CLINICAL SOURCES	
	4.3 DATA COLLECTION	
	4.4 DATA STATISTICS	
	4.5 DATASET CREATION	
	4.6 EXPERIMENTAL ENVIRONMENT	
5.	METHODOLOGY	25
	5.1 VQAUTISM AND BASELINE MODELS	
	5.2 OVERALL FLOW AND PIPELINE FOR VQAUTISM	
6.	RESULTS AND DISCUSSION	33
7.	CONCLUSION AND FUTURE WORK	38
	7.1 CONCLUSION	
	7.2 FUTURE WORK	
	REFERENCES/BIBLIOGRAPHY	42

LIST OF FIGURES

Figure No.	Title	Page No.
1.	ASD children to the left and TD children to the right.	14
2.	Line graphs for Youtube Dataset. Red line represents Typically Developed And the blue line depicts Autistic Children	18
3.	Line graphs for the Clinical Sources. Red line represents Typically Developed and the blue line depicts Autistic Children	19
4.	VQAutism Model Architecture	27
5.	Pipeline for the overall study	28
6.	Graph illustrating accuracy of Clinical Dataset	34
7.	Graph illustrating accuracy of Youtube Dataset	34
8.	Graph illustrating AUC-ROC of Youtube Dataset	35
9.	Graph illustrating AUC-ROC of Clinical Dataset	35

LIST OF TABLES

Table No.	Title	Page No.
1.	Thresholds for dataset creation	20-21
2.	Questions with their corresponding AUs	22
3.	Questions with their respective entropies and weights	31
4.	Accuracy of various video classification models on Clinical and Youtube datasets	32
5.	AUC-ROC metrics for ANN, Transformers and VQAutism	36

CHAPTER-1

INTRODUCTION

Psychological disorders are a group of mental health conditions that affect a person's thoughts, emotions, and behaviour. These disorders can range from mild to severe and can interfere with a person's ability to function in their daily life.

1.1 What is VQA?

VQA (Visual Question Answering) aims develop models that can understand and reason about the visual content of images or videos in the same way that humans do. Introducing VQA into psychology can make this research stand out. Recent years have seen vast progress in the field of Mental Health and Therapy.

- Online Therapy & Counselling
- Mental Assessment Apps
- Multiple Startups focused on Mental Health and well-being.
- 100+ of research articles have been published in 2022 alone

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental disorders that can have significant impacts on academic, social, and occupational functioning. Early diagnosis and intervention are critical for improving outcomes for individuals with these disorders. However, traditional methods of diagnosing these disorders rely on subjective observations and expert opinions, which can be time-consuming and prone to errors.

1.2 How can we address these disorders?

Analyse the video and audio of patients with a particular disorder, extract features and be able to confidently predict the disease using a novel VQA architecture. These include Facial Features, Spoken Words, Voice Modulation and Emotion Detection. These recognized patterns will then be able to be classified into a specific type of Psychological disorder.

We believe that a VQA focusing on Psychology can bring a massive impact in the field of Mental Wellbeing and Therapy. Further, this research could also develop into a fully fledged system to assist in the detection of Psychological illnesses.

VQA can aid in the early detection and objective assessment of these disorders by analysing videos of individuals with the disorders and identifying patterns associated with the disorders. Video datasets provide a rich source of behavioural information that can be used to identify markers associated with these disorders. VQA algorithms can extract relevant information from these videos, including body language, eye movements, facial expressions, and speech patterns, to identify patterns associated with these disorders.

The need for studying the application of VQA in the context of ASD arises from the need for early detection, objective assessment, identification of behavioural markers, and development of personalized interventions. By utilizing VQA, clinicians and researchers can improve the accuracy and reliability of diagnoses, develop more effective interventions, and ultimately improve outcomes for individuals with these disorders.

1.3 Applications:

- Improved assessment and diagnosis: By analysing various cues such as facial expressions, voice modulation, and body language, VQA can provide an improved

assessment of the presence and severity of symptoms.

- Remote monitoring and support: VQA can enable remote monitoring and support for individuals with psychological disorders. This can be especially helpful for individuals who have limited access to in-person care or who require ongoing support.
- Video analysis in medical instructions for healthcare workers and students in rural areas for the initial diagnosis of such diseases.

CHAPTER-2

PROBLEM STATEMENT

To build a Goal Oriented - VQA system that takes as input an VIDEO of a clinical interview and a NATURAL LANGUAGE QUESTION about the video and generates a natural language answer as the output along with the probability prediction of Autism Spectrum Disorder.

2.1 Constraints:

1. Data Availability
2. Psychology Domain Experts
3. Questionnaire Sets

2.2 Assumptions:

Extracted features from data are representative of the patient's behaviour and can be used to differentiate between psychological disorders.

2.3 Dependencies:

1. Dataset Source
2. Clinical Validation

2.4 Risks:

1. Unexplored Domain of Research
2. Ethical considerations of datasets
3. The VQA system's accuracy and performance may not be satisfactory, and there may be issues in predicting the psychological disorder correctly.

2.5 Objective of Study:

The objective of the research is to build a Goal Oriented - VQA system that takes as input a VIDEO and a NATURAL LANGUAGE QUESTION about the video and generates a natural language answer as the output. A VQA-assisted model for predicting Autism Spectrum Disorder by extracting facial features : Action Units (AUs), facial coordinates and emotions.

The system analyses the video and audio of patients with a particular disorder, extracts features, and confidently predicts the presence of the disease using a novel VQA architecture. The extracted patterns will then be classified into a specific type of psychological disorder.

CHAPTER-3

LITERATURE SURVEY

"Analysis and Detection of Autism Spectrum Disorder Using Machine Learning Techniques," a study by Suman Raja et al. [1], describes an effort to look at the potential application of Naive Bayes Algorithm, Support Vector Machines, Logistic Regressions, KNN, Neural Network, and CNN to the analysis and prediction of ASD issues in children, and adults. The second dataset for adult ASD screening has seven hundred and four cases and twenty-one attributes. For the third dataset used to screen teenagers for ASD, there are 104 cases and 21 attributes. Results clearly show that, after implementing multiple machine learning approaches and addressing missing values, CNN-based prediction models outperform all other models on these datasets, with a greater accuracy of 99.53%.

Combining a CNN with transfer learning and flask to create a deep learning application for detecting autism using facial features that have been tested in experiments., the study by Zeyad A. T. Ahmed et al. [2] aims to assist families and psychiatrists in making an accurate diagnosis of autism using a simple technique. The 3,014 facial images of a diverse group of children, including 1,507 autistic children and 1,507 non-autistic children, were taken from a publicly available dataset on Kaggle. MobileNet achieved an accuracy of 95%, Xception 94%, and InceptionV3 0.89% based on the classification results for the validation data.

SchiNet is a novel neural network architecture that Mina Bishay et al. [3] propose that estimates expression-related symptoms in two different assessment interviews. The proposed

SchiNet is assessed for the ability to predict schizophrenia symptoms without relying on a patient. According to experimental findings, some automatically recognised facial expressions are significantly correlated with schizophrenia symptoms, and the suggested network for determining symptom severity shows promise.

Shashank Jaiswal presents a dynamic deep learning system for accurate automatic face expression recognition [4]. It has been shown that the suggested system could be useful, impartial, and time-saving for the clinical diagnosis of ADHD and ASD.

In order to automate and streamline the process of determining the severity of emotion-related symptoms of schizophrenia, Debsubhra Chakraborty et al. [5] suggest using low-level acoustic signals to anticipate the expert subjective ratings given by a trained psychologist during an interview with the patient. Using machine learning algorithms and the leave-one-out cross-validation technique, the subjective ratings could be predicted with an accuracy of 61-85% from the objective openSMILE acoustic signals.

“The Potential of Accelerating Early Detection of Autism through Content Analysis of YouTube Videos”, a study by Vincent A. Fusaro et al. [6], explores the possibility of using YouTube videos to detect early signs of autism in children. The paper notes that early detection of autism is critical for effective intervention and that current diagnostic methods are often time-consuming and costly. The authors propose a method of using machine learning algorithms to analyse YouTube videos of children to detect early signs of autism. They used machine learning algorithms to analyse the videos for signs of autism, such as lack of eye contact, repetitive behaviours, and delayed speech. Similar explorations are done on various datasets like home videos recorded on mobile devices [7].

"Automatic Detection of ADHD and ASD from Expressive Behaviour in RGBD Data", by Shashank Jaiswal et al. [8] explores the use of machine learning algorithms to detect early signs of ADHD and ASD from expressive behaviour in RGBD data. The authors collected a dataset of over 800 videos of children between the ages of 4 and 12 and used machine learning algorithms to analyse the videos for signs of the disorders, such as hyperactivity, inattention, and social interaction. The results show that it is possible to detect signs of ADHD and ASD from RGBD data with a reasonable level of accuracy, providing a promising approach to early detection and intervention for these disorders. Further using symptoms like eye Movements different models like DeepGaze II with VGG-19 features and Gaussian kernel with SWAN Scores were tried out [9].

"Single and Cross-Disorder Detection for Autism and Schizophrenia" by Aleksander Wawer et. al [10] uses Cross dataset approaches and few shot learning methods to detect schizophrenia and ASD. The authors used BOW, POS, XGBoost, Linear SVC and Neural Network to effectively detect the disorders. Similarly, gestures of patients are used to detect ADHD using Dynamic Time warping and a set of gesture samples of a certain gesture category using either Gaussian mixture models or an approximation of convex hulls [11].

"Atypical postural control can be detected via computer vision analysis in toddlers with autism spectrum disorder" by Geraldine Dawson et al. [12] uses Time series data for autism spectrum disorder. A Study[13] employed an objective assessment method to provide more accurate evaluations of therapeutic effects in 25 children with ADHD (23 boys and 2 girls). To evaluate the participants' improvement and treatment's effectiveness, the pixel subtraction technique was used in video analysis.

"Classification of Schizophrenia Data Using Support Vector Machine (SVM)" by T V Rampisela et. Al [14] aims to distinguish people who are Schizophrenics and people who are

not using Support Vector Machines. Further [15] uses Deep learning based system developed to detect the Autism, ADHD, Schizophrenia disorders.

“A novel automated autism spectrum disorder detection system” by Bing-Jhang Lin et. [16] developed a “unique classification system to categorize autism and normal EEG signals”. Eighteen nonlinear features were extracted from the acquired brain signals. “Mental Status Detection for Schizophrenia Patients via Deep Visual Perception” by Bing-Jhang Lin et. al. [17]. Further [18-20] talks about multiple DL methods on Autism, ADHD multiple types of data, very specifically about the challenges and gaps in the related research.

“ADHD Intelligent Auxiliary Diagnosis System Based on Multimodal Information Fusion” by Yanyi Zhang [21], implemented a new system to detect and capture multimodal information and predict psychological behaviour. VGGNet and CNN were used along with Hardware + Software functionalities integration. Further[22-24] uses incoherent speech(with coherence models and classifiers), Computer vision for facial behaviour analysis and various deep learning architectures for the detection of schizophrenia.

In the study “Voicing a Schizophrenic Mind: A natural language processing based approach to diagnose schizophrenia using speech” by Raghunathan, et.al, collected speech samples from patients with schizophrenia and healthy controls to develop a machine learning-based diagnostic tool. The results demonstrated that significant differences existed in the speech patterns of individuals with schizophrenia compared to healthy controls [25].

A Study[26] proposes a method to diagnose Autism Spectrum Disorder (ASD) using video games and deep learning techniques to analyse facial expressions. The authors collect facial expression data from children with ASD and healthy controls while playing a video game, and use this data to train a deep learning model. Lee, Jung Hyuk, et al in the study

“Deep-learning-based detection of infants with autism spectrum disorder using auto-encoder feature representation” proposes a deep learning-based approach for detecting Autism Spectrum Disorder (ASD) in infants using auto-encoder feature representation [27]. The authors collect electroencephalography (EEG) data from infants with and without ASD and use it to train a deep learning model. A Study[28] proposes a deep learning-based technology for classifying Attention-Deficit/Hyperactivity Disorder (ADHD) and ADHD risk in children by recognizing their abnormal behaviours during a robot-led ADHD screening game. The authors collect video data of children playing the game and use it to train a deep learning model to identify abnormal behaviours that are associated with ADHD.

A machine learning-based approach is proposed in [29] for predicting Attention-Deficit/Hyperactivity Disorder (ADHD) in children using behavioural activity data. The authors use accelerometer data to train a model that accurately predicts children with ADHD based on their activity patterns. An automatic recognition system for schizophrenia from facial videos is proposed in [30] using a 3D convolutional neural network (CNN). The authors train a deep learning model using facial videos from patients with schizophrenia and healthy controls, which accurately classifies individuals with schizophrenia and healthy controls based on their facial expressions.

In the study "Sch-net: A Deep Learning Architecture for Automatic Schizophrenia Selection." Fu, Jia, et al proposes Sch-net, a deep learning architecture proposed for automatic detection of schizophrenia using brain MRI data [31]. The model accurately classifies individuals with schizophrenia and healthy controls based on their MRI data, suggesting its potential as a diagnostic tool for schizophrenia.

In a study[32] the authors create TutorialVQA dataset by collecting and annotating video clips from tutorial videos on various topics, including cooking, crafts, and computer

software. They suggest that this dataset can be used to develop models that can automatically answer questions in tutorial videos, making them more accessible and useful for learners. [33] compares the performance of four different language transformers, BERT, RoBERTa, DistilBERT, and GPT-2, for video question answering (VQA) tasks. The study evaluates their performance on two VQA datasets and analyses the impact of model size, pre-training task, and fine-tuning on their performance. The results show that RoBERTa and GPT-2 outperform BERT and DistilBERT, highlighting the importance of pre-training tasks and model size for VQA.

The paper [34] presents the KnowIT VQA dataset and task for answering knowledge-based questions about videos. The dataset includes video clips and questions from various domains, requiring both video content understanding and general knowledge. The authors propose a method for jointly encoding video frames and text using a pre-trained visual-linguistic transformer and demonstrate its potential for answering knowledge-based questions. The proposed method and dataset can be useful for educational and informational purposes.

The paper “Modern question answering datasets and benchmarks: A survey” by Wang, Zhen [35] surveys 45 datasets and benchmarks for modern question answering (QA) tasks, including open-domain QA, knowledge-based QA, visual QA, and multi-modal QA. The authors analyse the characteristics of each dataset, identify common challenges and limitations, and provide a resource for researchers and practitioners to select appropriate datasets for their work.

A Study [36] proposes V-Doc, a dataset and task for answering visual questions with documents. The authors combine text documents and images from the OpenImages and Wikipedia datasets, generate questions that require multi-modal understanding, and evaluate a baseline model using a pre-trained visual feature extractor and language model. The paper

“Mmbert: Multimodal bert pretraining for improved medical vqa” by Khare et al. presents Mmbert, a multimodal BERT-based pretraining method for medical visual question answering [37]. The authors fine-tune BERT on a dataset of medical images and corresponding questions and answers, incorporating medical text data to improve pretraining. They demonstrate that Mmbert outperforms several state-of-the-art methods on a benchmark dataset for medical VQA, showing that incorporating medical text data can improve the performance of pretraining models for medical VQA tasks.

A Study[38] proposes TGOD, a Text-Guided Object Detector for multi-modal video question answering. The method uses a pre-trained object detection model and a language model to detect relevant objects in video frames and generate textual descriptions, which guide the attention mechanism of the VQA model. Further [39] proposes a Pyramidal Multimodal Transformer (PMT) model for efficient video question answering, which uses a pyramidal architecture combining a convolutional neural network and a transformer-based architecture to capture high-level and low-level visual features. The model integrates audio and textual embeddings to enhance its performance and outperforms several state-of-the-art models while being computationally efficient.

A Study [40] presents a new Medical Instructional Video Question Answering (MIVA-QA) dataset for medical instructional video classification and question answering, which contains 100 videos with annotations for relevant video segments. In "Visual Question Answering on 360° Images," the authors propose a VQA model that can process and answer questions about 360-degree images by transforming them into 2D images using a panoramic projection, and using a CNN and LSTM to extract visual features and process sequential information [41]. The paper shows that their model outperforms several state-of-the-art models on this task, and has potential applications in virtual reality, tourism, and entertainment.

In "xGQA: Cross-lingual visual question answering," the authors develop a cross-lingual VQA model that can answer questions in multiple languages using a shared visual-semantic space [42]. They employ a transformer-based architecture and a shared embedding space for visual and textual features. The paper shows that their model outperforms several state-of-the-art models on cross-lingual VQA, and has potential applications in machine translation, image captioning, and video surveillance.

A Study [43] introduces a new dataset called DocVQA for visual question answering on document images. The dataset consists of 5,000 images with 72,000 questions that require reasoning about text, images, and layout. The authors evaluate several state-of-the-art VQA models on the dataset and show that it poses significant challenges for these models, highlighting the need for further research in this area. Further [44] proposes a Dual-Visual Graph Reasoning unit for video question answering, which integrates spatial and temporal reasoning to improve performance on this task. The authors evaluate their method on several benchmarks and show that their model outperforms several state-of-the-art models on these tasks.

CHAPTER-4

DATA

To comprehensively address the scarcity of publicly available video datasets, this research necessitated a meticulous curation process encompassing the compilation of video footage from diverse sources. In order to construct robust and representative datasets, the collection strategy employed a dual-pronged approach, drawing from both widely accessible online platforms and specialized clinical sources.

4.1 YouTube Videos:

The research involved a compilation of videos obtained from YouTube featuring recordings of children with Autism Spectrum Disorder (ASD) and typically developing (TD) children. These selected videos were curated and utilized for the research, offering a wide array of diversity and interactions involving children from both groups. Each category comprised a careful curation - including 18 videos of ASD children and 18 videos of TD children. Figure 2. Shows the sample of the dataset collected.

4.2 Clinical Sources:

Video footage of children with ASD was recorded at Bangalore Medical College and Research Institute (BMCRI) and the Academy of Severe Handicaps and Autism (ASHA). This deliberate choice of settings aimed to provide a comprehensive understanding of behaviours exhibited by children with ASD with specific clinical contexts. These videos showcase a spectrum of behaviours, interactions, and emotional responses, offering valuable insights into the nuanced aspects of their experiences of children with ASD in a clinical

setting. The footage emphasizes the naturalistic behaviours, interactions and emotional responses of the children within the clinical setting.

Conversely, video footages featuring typically developing (TD) children was sourced from a government school in Udupi, Karnataka. The decision to capture these videos in a school setting aligns closely with everyday community environments, providing a window into the typical behaviours, social interactions, and emotional responses of TD children within the context of a regular educational setting. This choice enhances the ecological validity of the dataset, ensuring that the behaviours observed are reflective of the natural dynamics of children in community-based settings.



Figure 1. ASD children to the left and TD children to the right.

4.3 Data Collection:

The acquisition of data for this study underwent a rigorous and systematic process. Initial groundwork involved the administration of a detailed survey to parents or guardians of both children diagnosed with Autism Spectrum Disorder (ASD) and typically developing

(TD) children. The survey was methodically designed to probe into various facets, including the nature of digital content consumption, viewing durations, attention spans, preferred genres, and specific preferences and aversions, among other pertinent factors. This comprehensive survey served as a foundational element, providing nuanced insights into the media consumption patterns of the subjects. Subsequent to the meticulous analysis of the survey data and in adherence to research requirements, a discerning selection process identified participants from both the ASD and TD cohorts. Stringent ethical standards were upheld as informed consent was diligently obtained from the parents or guardians of each participating child.

To standardize the stimulus for emotional response analysis, a singular 6-minute video was meticulously curated. This compilation integrated footage from a diverse array of sources, a process informed by the survey insights pertaining to content types and preferences. The video content was thoughtfully structured, featuring scenes deliberately designed to evoke specific emotions such as happiness, sadness, anger, and others. Each participant, irrespective of their diagnostic category, was then instructed to view the entire 6-minute video, while a camera methodically captured their emotional responses.

4.4 Data Statistics:

The overall video dataset encompasses a meticulous selection of 12 videos portraying children diagnosed with Autism Spectrum Disorder (ASD). These recordings were conducted in a clinical setting at Bangalore Medical College and Research Institute (BMCRI) and the Academy of Severe Handicaps and Autism (ASHA). The ASD subgroup comprises 8 boys and 4 girls. Additionally, the dataset includes 12 videos featuring typically developing (TD) children sourced from a Government school in Udupi, India. This group consists of 6 boys and 6 girls.

Both the ASD and TD datasets are characterized by children within the 5 to 7 years age range. The decision to focus on this specific age group was guided by clinical suggestions and consultations, with an emphasis on early intervention. The selected age range aligns with critical developmental stages, allowing for a nuanced exploration of behaviors and interactions during a pivotal period of childhood. This choice reinforces the research's commitment to capturing early manifestations of behaviors in children with ASD and their typically developing counterparts, contributing to the overarching objective of advancing understanding and potential early interventions in the context of ASD.

4.5 Dataset Creation:

1.Extracting Features: Facial expressions help in giving a glimpse into a person's inner mental state and serve as nonverbal cues that assist in communication among individuals. A significant hurdle in studying facial expressions has been finding the most effective methods to accurately represent and measure these expressions objectively. The Facial Action Coding System (FACS) is one of the widely accepted systems used to precisely measure the strength of specific groups of facial muscles known as Action Units (AUs). Tools like OpenFace and Py-Feat were used for this process. OpenFace is a state-of-the-art tool designed for facial landmark detection, head poses estimation, facial action unit recognition, and eye-gaze estimation. Py-Feat is an open-source Python toolbox that provides support for detecting, preprocessing, analyzing and visualizing facial expression data. Each video is processed to extract image frames. These frames were utilized as input for the Py-Feat toolkit to extract various facial expressions, action units, emotions, eye gaze, and face pose. Some of the features to mention include FaceRectX, FaceRectY, x_0, x_1, y_0, y_1, z_0, z_1, AU1,

AU43, anger, happiness, etc. This process of feature extraction was implemented onto every single video that was collected from both the sources - YouTube and Clinical sources.

2. Questions: These datasets obtained from both YouTube and clinical sources have questions categorized into 11 categories: reduced eye contact, puffed or raised cheeks, horizontally furrowed inner brows, parted lips, pursed lips, excessive smiling, teeth grinding, atypical gaze, lip corner depressor, asymmetric face and inner brow raiser. All the answers were close-ended (yes/no). All the questions were validated and recommended by certified medical practitioners specialising in psychiatry for this study. The questions have proven to be significant in identifying autism through facial expressions.

Leveraging VQA to Analyse Autism Spectrum Disorder (ASD)

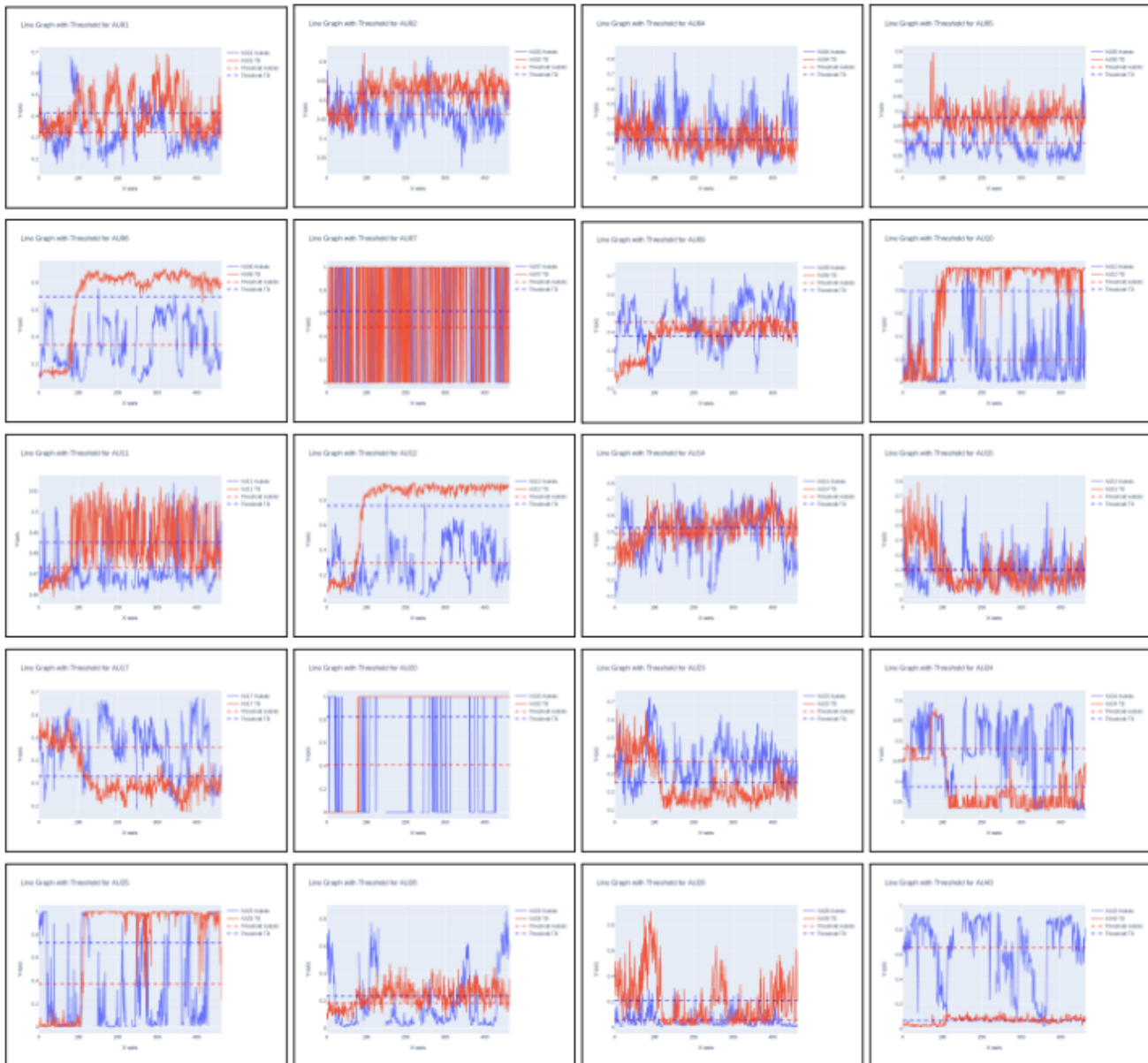


Figure 2. Line graphs for Youtube Dataset. Red line represents Typically Developed and the blue line depicts Autistic Children

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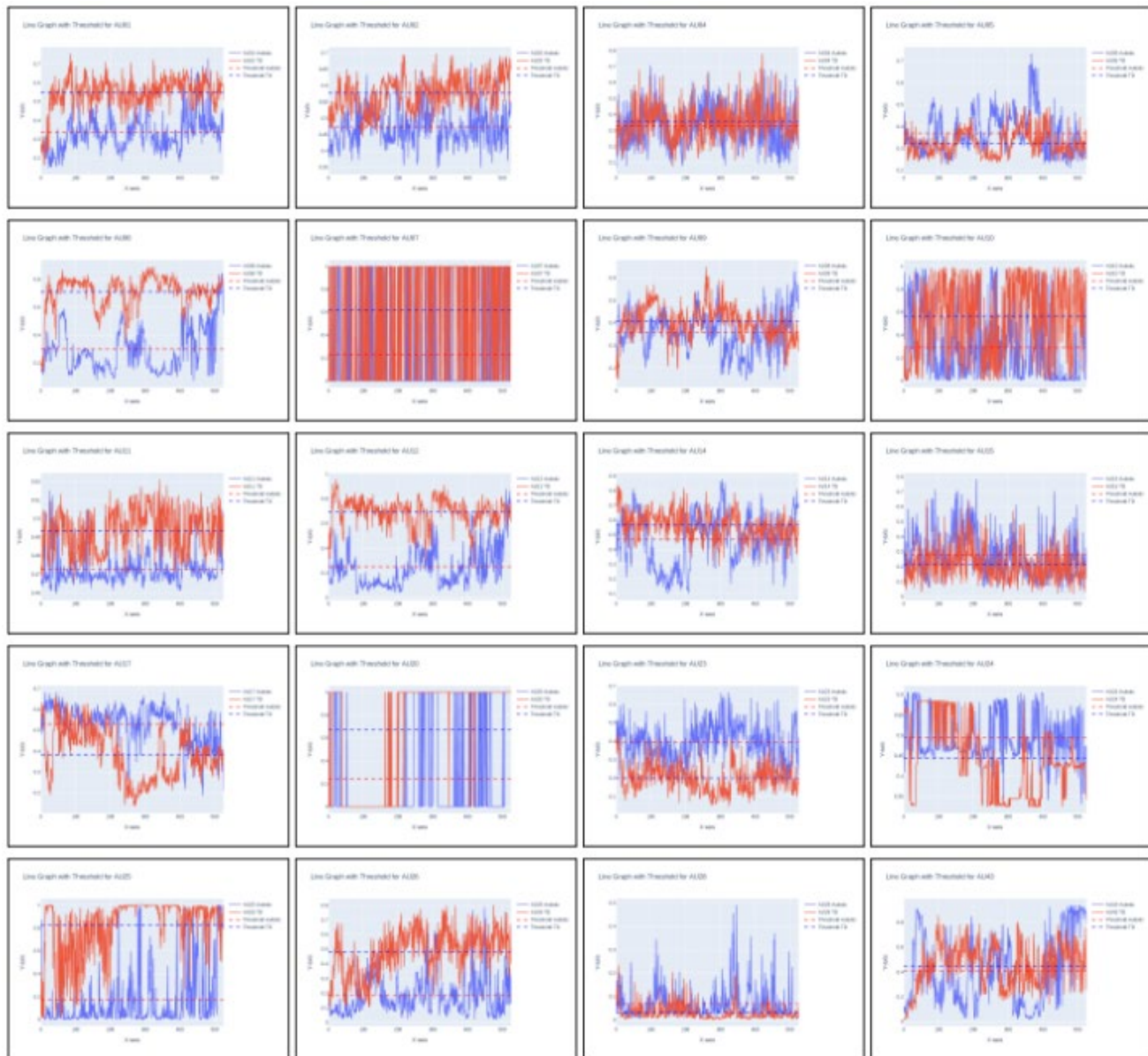


Figure 3. Line graphs for the Clinical Sources. Red line represents Typically Developed and the blue line depicts Autistic Children

3. **Generating Answers:** Answers to these questions were generated by applying specific thresholds to the action units associated with the corresponding behavioural aspects and facial expressions of the particular question. These thresholds were categorized into 3 distinct types - Manual Thresholds (validated by medical counsel), mean-based thresholds, and mode-based thresholds. If these values exceed the thresholds they are classified as a "yes"; otherwise they are categorized as a "no." Manual thresholds have different levels and these can be seen in Table 1. These three datasets provide diverse categories for evaluating and training machine learning models.

Table 1. Thresholds for dataset creation

Action Unit	YouTube Dataset			Clinical Sources		
	Manual Threshold	Mean	Mode	Manual Threshold	Mean	Mode
AU01	> 0.45	0.408300	0.388235	> 0.45	> 0.486410	> 0.491667
AU02	-	0.503191	0.500000	-	> 0.403275	> 0.366667
AU04	> 0.4	0.335578	0.300000	> 0.4	> 0.323611	> 0.216667
AU05	-	0.451971	0.464706	-	> 0.381392	> 0.35
AU06	< 0.5	0.377280	0.358824	< 0.5	> 0.214852	> 0.191667
AU07	-	0.398422	0.294118	-	> 0.128631	-
AU09	$0.2 < AU9 < 0.6$	0.333077	0.317647	$0.2 < AU9 < 0.6$	>	> 0.15

					0.252831	
AU10	-	0.374367	0.294118	-	> 0.156180	> 0.083333
AU11	-	0.476364	0.500000	-	> 0.552240	> 0.75
AU12	< 0.5	0.399736	0.364706	< 0.5	> 0.177430	> 0.141667
AU14	-	0.408843	0.382353	-	> 0.393074	> 0.366667
AU15	0.01 < AU15 < 0.3	0.236664	0.164706	0.01 < AU15 < 0.3	> 0.409363	> 0.4
AU17	-	0.480352	0.494118	-	> 0.510736	> 0.525000
AU20	0	0.549492	0.529412	0	> 0.238877	> 0.083333
AU23	< 0.5	0.338296	0.317647	< 0.5	> 0.506488	> 0.483333
AU24	0.3 < AU24 < 0.6	0.464782	0.488235	0.3 < AU24 < 0.6	> 0.539772	> 0.558333
AU25	0.2 < AU25 < 0.5	0.530977	0.535294	0.2 < AU25 < 0.5	> 0.406048	> 0.333333
AU26	0.2 < AU25 < 0.5	0.315380	0.235294	0.2 < AU25 < 0.5	> 0.238633	> 0.15
AU28	-	0.115279	0.058824	-	> 0.238381	> 0.133333
AU43	>0.2	0.179683	0.100000	>0.2	> 0.259862	> 0.216667

Table 2. Questions with their corresponding AUs

Question	Action Units
Are there signs of Unusual/ excessive smiling?	AU6, AU12, AU15, AU17, AU23, AU24, AU25, AU26
Is the face asymmetric (FA)?	AU6, AU9, AU43
Are their lips parted?	AU25, AU 26
Are cheeks puffed or raised with happiness?	AU6
Are there any noticeable lip movements, (such as lip pursing, biting, or smacking, exhibited by the child in the video)?	AU20
Does the lip corner depressor show discomfort or stress?	AU15
Are there signs of jaw clenching or teeth grinding?	AU25, AU26
Does the child have reduced eye contact?	AU43, AU6
Does the child have atypical gaze patterns?	AU43
Are the inner portions of the brows furrowed horizontally?	AU4
Is there absence of the inner brow raiser?	AU1

4.6 Experimental Environment:

In this research endeavor, Python version 3.10.11 served as the programming foundation for implementing various machine learning techniques. Google Colab provided the computational environment for conducting tests, offering a scalable and collaborative platform. The implementation leveraged prominent deep learning frameworks, namely PyTorch, TensorFlow, and Keras, to harness cutting-edge methodologies and algorithms in

the pursuit of effective results. We intend to publish both the research findings and open-source datasets on GitHub. It is noteworthy that clinically procured data, held to stringent privacy and ethical standards, will not be disclosed to maintain the confidentiality and integrity of sensitive information.

CHAPTER-5

METHODOLOGY

Traditional question answering (QA) models often fall short in capturing both textual and emotional features. This work bridges this gap by proposing an advanced bi-modal model that unifies textual and facial information through a deep neural network architecture. Various bi-modal models were tried and tested.

5.1 VQAutism and Baseline Models:

1. Artificial Neural Network:

Textual questions are tokenized using a tokenizer, and sequences are padded to ensure uniform input length. The dataset comprises questions, answers, and facial expression features extracted from videos. Facial expression features include Action Units (AUs). Textual data is tokenized. The tokenized sequences are then embedded using a word embedding layer with a dimensionality of 100. The embedding layer transforms words into dense vectors, capturing semantic relationships. The embedded text sequences are flattened, and the facial expression features are concatenated with the flattened text embeddings. The concatenated features are fed into a simple artificial neural network architecture. A single dense layer with a sigmoid activation function is employed as the output layer. The simplicity of this architecture allows for efficient training and interpretation of learned representations.

While the simplicity of the Artificial Neural Network (ANN) architecture provides a straightforward approach to question answering, it comes with inherent limitations. The model may struggle with capturing sequential dependencies in textual data, handling variable-length inputs effectively, and extracting hierarchical features. Additionally, the model's basic embedding layer may limit its ability to grasp complex relationships within textual content, and its simplicity may hinder effective learning of long-term dependencies.

2. Transformers:

This model incorporates facial expression features into a BERT-based transformer model, enriching the contextual understanding for improved QA performance. Textual data is tokenized using the BERT-based transformer's tokenizer. The resulting tokenized sequences, along with attention masks, serve as inputs to the transformer model. The transformer embeddings are pooled, yielding a contextually rich representation of the input text. Facial expression features, encompassing Action Units (AUs) and emotion labels, are incorporated as additional input features. The neural network architecture consists of three main inputs: tokenized input IDs, attention masks, and facial expression features. The transformer embeddings are obtained from the BERT-based model, and a pooled representation is derived. This representation is concatenated with the facial expression features, forming a feature vector. A final dense layer with sigmoid activation produces binary predictions.

The proposed model demonstrates robust performance in question-answering tasks. The integration of BERT-based transformer embeddings and facial expression features enhances the model's understanding of context, resulting in improved contextualized responses. However, its computational complexity, interpretability challenges, and data requirements may pose limitations, especially in resource-constrained scenarios.

3. VQAutism

This work introduces an innovative model that combines advanced text processing using dual LSTM layers with facial expression features. The goal is to improve how question-answering systems handle both text intricacies and emotions, making them more effective and versatile. The dataset comprises textual questions, corresponding answers, and facial expression features. Tokenized textual questions are padded, while facial expression features encompass a range of Action Units (AUs).

The first LSTM layer captures sequential information from the embedded sequences, creating a rich foundation for context-aware understanding. The second LSTM layer refines this output, focusing on longer-term dependencies and unveiling deeper contextual representations. Facial expression features are integrated with the output of the second LSTM layer, producing a novel fused feature vector. This fusion represents an approach to question answering, combining textual and non-verbal information in a singular, comprehensive representation. This architectural choice empowers the model to distil patterns from the feature vector, contributing to its capabilities in capturing complex dependencies.

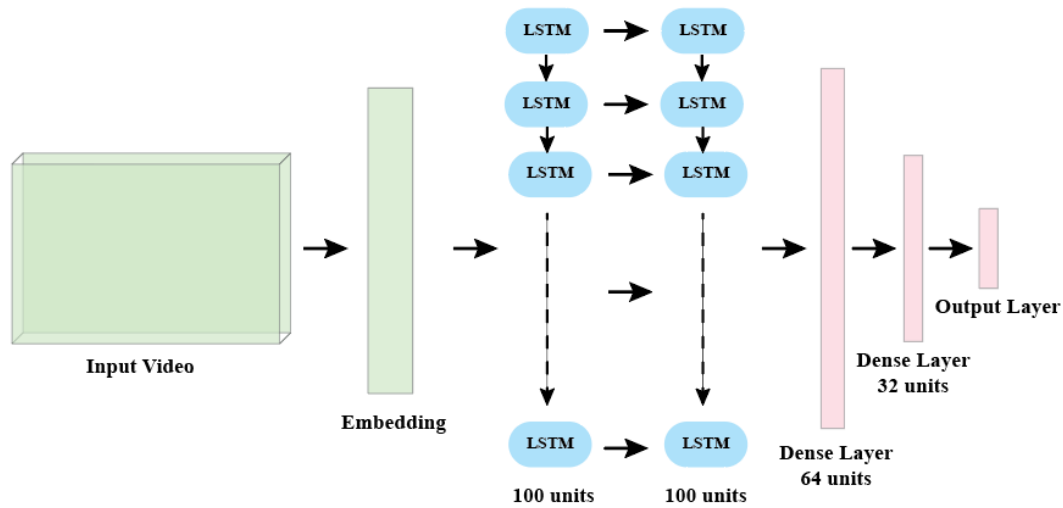


Figure 4. VQAutism Model Architecture

5.2 Overall Flow and Pipeline for VQAutism:

1. Overall Pipeline

The objective of the study is to detect the probability of ASD. To do so we follow the below steps in succession to get the desired probability.

- Frames are extracted from the input video that is passed.
- Extract features from each frame and store the corresponding action units.
- Ask relevant questions and pass the questions and the frames through the models.

- Get the predicted answers and pass them through the probability algorithm with the assigned weights for each question to predict the probability of ASD.

The pipeline for prediction of the probability of ASD using supervised learning algorithms is illustrated in Figure 5.

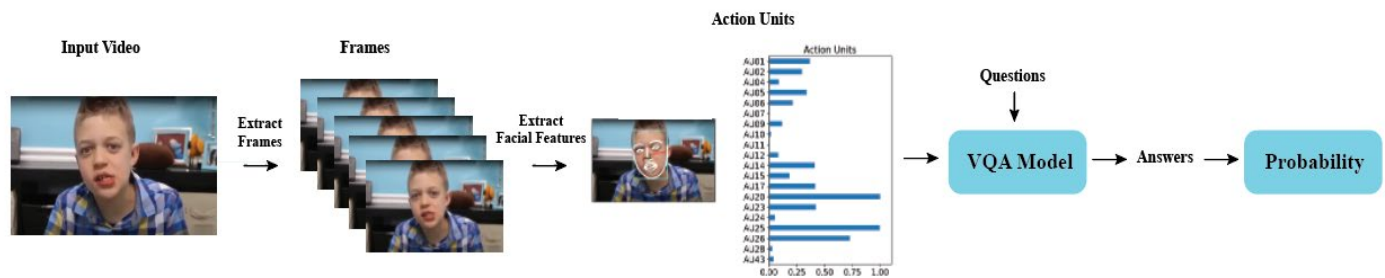


Figure 5. Pipeline for the overall study

2. Assigning weights

Information entropy indeed serves as a powerful tool in quantifying uncertainty or randomness within potential outcomes. Initially applied to quantify clinical diagnostic information, information theory has expanded its reach across various domains, illustrating its versatility and applicability [44].

Its utilization spans diverse fields such as molecular biology, psychology, quantum mechanics, music, and beyond. For instance, in molecular biology [45], it aids in analyzing genetic sequences and understanding the complexity of biological systems. In psychology [46], information theory contributes to understanding cognitive processes and decision-making. Even in quantum mechanics [47], it helps comprehend the uncertainty and

information associated with quantum states. Additionally, in the realm of music [48], information theory facilitates the analysis of musical patterns and compositions.

One critical aspect where information theory plays a pivotal role is in addressing fundamental inquiries, such as quantifying the information expected from the diagnosis of an individual's state [49]. This application highlights its significance in healthcare and clinical settings, enabling the quantification of information received from diagnostic data. By measuring the information content inherent in diagnostic outcomes, information theory aids in assessing uncertainty, guiding decision-making, and providing a quantitative framework for understanding the complexity of diagnostic information. The ability of information theory to provide a quantitative measure of information content is instrumental across these diverse domains, offering a unified framework to quantify uncertainty and information content, thereby enabling deeper insights and informed decision-making processes.

In this research, information entropy has been strategically employed to interpret and allocate weights to questions within the VQA system. The binary nature of the answers, i.e., yes or no, transforms each question into a random variable, and the aim is to capture and understand the uncertainty associated with these variables.

The assignment of weights to each question is intricately linked to its entropy. Entropy, in this case, serves as a quantifiable metric of uncertainty. The higher the entropy, the greater the uncertainty associated with the question; conversely, lower entropy implies a higher degree of predictability.

Formula used:

$$H = - \sum p(x) \log p(x)$$

(1)

In Table 3, the assigned weights for each question are determined by subtracting the entropy from 1. This approach aligns with the principle that higher weights correspond to questions that contribute more valuable information to the system. The rationale behind this lies in the inverse relationship between entropy and predictability. Questions with higher entropy, indicative of greater uncertainty, are assigned higher weights, signifying their potential to yield more informative outcomes.

By incorporating information entropy in this manner, the research methodology not only acknowledges the varying degrees of uncertainty associated with different questions but also emphasizes the importance of leveraging this uncertainty as a guide for assigning weights. This approach enhances the efficiency of the VQA system, ensuring that questions with greater potential for information gain are accorded higher significance in the learning process.

Table 3: Questions with their respective entropies and weights

QUESTION	YES	NO	ENTROPY	WEIGHT
Does the child have reduced eye contact?	201	253	0.990515967	0.009484033
Are cheeks puffed or raised with happiness?	398	55	0.533408337	0.466591663
Are the inner portions of the brows furrowed horizontally?	218	235	0.998983873	0.001016127
Are their lips parted?	332	122	0.839625478	0.160374522
Are there any noticeable lip movements, (such as lip pursing, biting, or smacking, exhibited by the child in the video) ?	321	132	0.870510973	0.129489027
Are there signs of Unusual/ excessive smiling?	327	127	0.85509041	0.14490959
Are there signs of jaw clenching or teeth grinding?	331	122	0.840480956	0.159519044
Does the child have atypical gaze patterns?	200	253	0.990103195	0.009896805
Does the lip corner depressor show discomfort or stress?	367	86	0.701148093	0.298851907
Is the face asymmetric(FA)?	207	247	0.994393187	0.005606813
Is there absence of the inner brow raiser?	252	201	0.990837592	0.009162408

CHAPTER-6

RESULTS AND DISCUSSIONS

Table 4 shows the accuracy of three video classification models on two different datasets. The models are VQAutism, BERT, and ANN. The datasets are Clinical and Youtube.

Table 4. Accuracy of various video classification models on Clinical and Youtube datasets

Model	Dataset	Threshold	Mean	Mode
ANN	Youtube	76.91	73.90	72.98
	Clinical	82.76	76.44	75.07
Transformer	Youtube	66.53	64.26	62.29
	Clinical	78.50	71.98	65.99
VQAutism	Youtube	98.90	98.88	98.83
	Clinical	99.26	99.22	99.01

The empirical evaluation of the three distinct models—VQAutism, the BERT-based transformer model, and the simpler Artificial Neural Network (ANN)—reveals notable

variations in their performance within the context of the question-answering task. VQAutism emerges as the top-performing model, showcasing superior accuracy. Its effectiveness lies in its architecture, seamlessly integrating textual and facial information, which proves particularly advantageous for tasks such as question answering. The BERT-based transformer model demonstrates decent performance, leveraging advanced contextual understanding, but its computational complexity and resource demands position it below VQAutism. Conversely, the simpler ANN exhibits commendable performance, especially in scenarios with limited computational resources and data, however, with compromises in contextual understanding. These results underscore the interplay between model complexity, interpretability, and performance, offering insights into their respective strengths and weaknesses for question-answering applications.

VQAutism's performance on the Clinical dataset is particularly noteworthy. This dataset is challenging because it contains videos of patients with a variety of different medical conditions. VQAutism's ability to accurately classify these videos suggests that it could be used to develop new clinical decision support systems that can help doctors diagnose and treat their patients more effectively. Overall, the results of this study suggest that VQAutism is a promising new video classification model with the potential to be used in a variety of different applications.

In Figure 6, the graph depicts the accuracy of the models applied to the Clinical Dataset, showcasing its performance in terms of correct predictions. Figure 7 similarly illustrates the accuracy but for the Youtube Dataset, providing insights into the model's effectiveness in a different domain. Figures 8 and 9 specifically focus on the AUC-ROC, offering a comprehensive view of the model's discrimination ability for both Youtube and Clinical Datasets, respectively.



Figure 6. Graph illustrating accuracy of Clinical Dataset

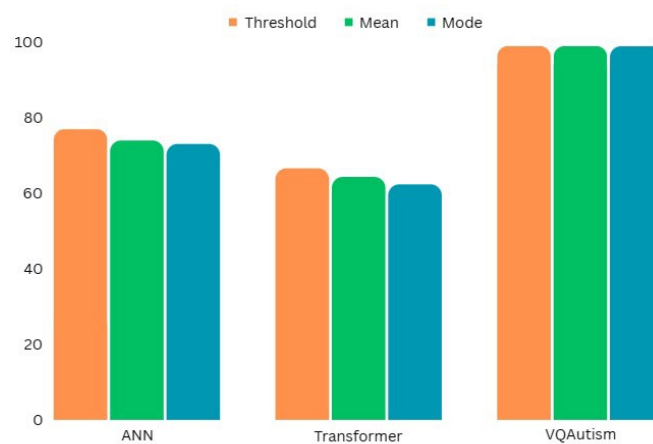


Figure 7. Graph illustrating accuracy of Youtube Dataset

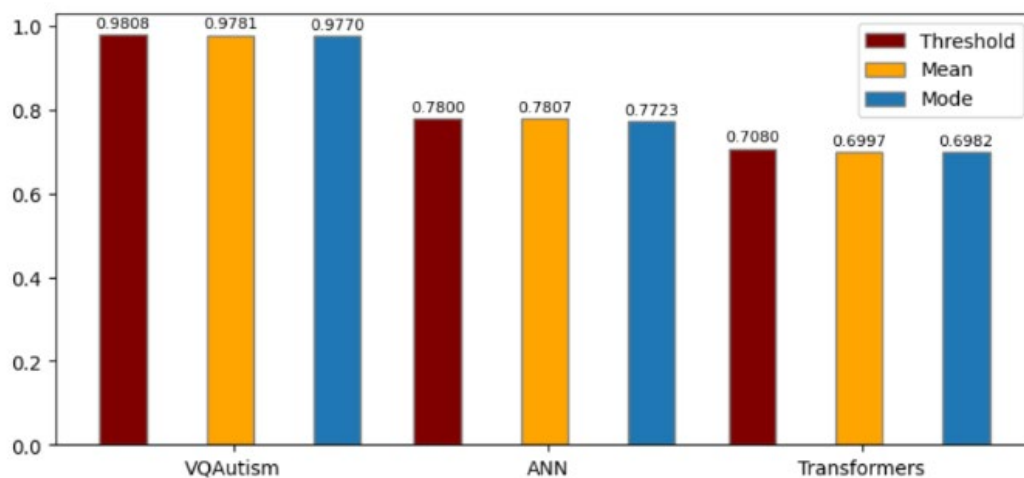


Figure 8. Graph illustrating AUC-ROC of Youtube Dataset

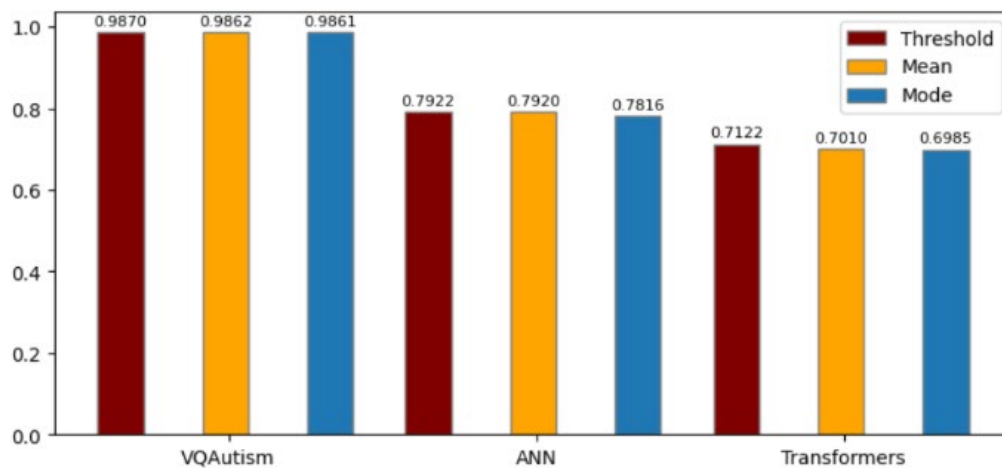


Figure 9. Graph illustrating AUC-ROC of Clinical Dataset

Table 5. AUC-ROC metrics for ANN, Transformers and VQAutism

Model	Type of Dataset	Parameter	Auc roc	Test loss	Test Accuracy	Test auc roc
VQAutism	Youtube	Threshold	0.9808	0.0495	0.9836	0.9980
		Mean	0.9781	0.0614	0.9741	0.9970
		Mode	0.9770	0.0615	0.9770	0.9915
	Clinical	Threshold	0.9870	0.0128	0.9856	0.9984
		Mean	0.9862	0.0125	0.9853	0.9910
		Mode	0.9861	0.0480	0.9865	0.9878
ANN	Youtube	Threshold	0.7800	0.4955	0.8115	0.8010
		Mean	0.7807	0.4200	0.8121	0.8002
		Mode	0.7723	0.5148	0.7985	0.7955
	Clinical	Threshold	0.7922	0.4950	0.7925	0.7951
		Mean	0.7920	0.5010	0.7955	0.7925
		Mode	0.7816	0.5484	0.7910	0.7914
Transformers	Youtube	Threshold	0.7080	0.5116	0.7200	0.7050
		Mean	0.6997	0.5201	0.7122	0.6987
		Mode	0.6982	0.5195	0.7015	0.6992
	Clinical	Threshold	0.7122	0.5048	0.7100	0.7105
		Mean	0.7010	0.5011	0.7025	0.6932
		Mode	0.6985	0.5240	0.7010	0.6988

CHAPTER-7

CONCLUSION AND FUTURE WORK

7.1 Conclusion:

This work endeavors to address the complex task of predicting Autism Spectrum Disorder (ASD) through a Goal-Oriented Visual Question Answering (VQA) system. The study explores three distinct computational techniques, namely Artificial Neural Network (ANN), Transformers, and the innovative VQAutism model, each capturing textual and emotional features from clinical interview videos. The proposed pipeline integrates video frame extraction, feature extraction, and question-answering models to predict the probability of ASD. Results and discussions underscore the performance of each model, with VQAutism emerging as the top-performing model, demonstrating superior accuracy, especially on the Clinical dataset. The study illuminates the trade-offs between model complexity, interpretability, and performance, providing valuable insights for future research and development in the domain. Ethical considerations, user-centric design, and longitudinal studies involving community engagement are emphasized to ensure responsible and trustworthy development of ASD diagnostic tools.

In summary, this work not only contributes to the ongoing discourse on leveraging artificial intelligence for mental health diagnostics but also outlines a roadmap for future that prioritize accuracy, reliability, and ethical considerations in the development of tools aimed at improving the early detection and intervention of Autism Spectrum Disorder. Through continued research and collaboration with the clinical community, the proposed system holds the potential to make a meaningful impact on the lives of individuals affected by ASD.

7.2 Future Work:

It is crucial to throw light on the future work of this research study to leverage the accuracy, reliability, and accessibility of diagnostic methods for the detection of Autism Spectrum Disorder (ASD). This could lead to an earlier and more accurate diagnosis of ASD, which is essential for effective intervention and could also help to identify children who may need further evaluation and decrease the number of missed diagnoses. Additionally, this research could lead to the development of new tools and resources for the early diagnosis and intervention of ASD, which could help to improve the long-term outcomes for children with ASD.

Enhancing Feature Extraction and Data Augmentation: Deep learning techniques, such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs), can be used to extract hierarchical features from video frames of facial expressions. Data augmentation techniques, such as rotation, scaling, and brightness modifications, can improve the diversity of the training dataset and help the model to adapt to various variances in emotions and facial expressions. Future research can focus on improving the process of feature extraction by investigating more sophisticated deep learning-based methods that are capable of extracting more discriminative and finely detailed features from video footage. Additionally, data augmentation techniques can be explored to enhance the performance of the model, especially when working with small amounts of labeled data. These improvements in feature engineering can lead to more accurate predictions and better generalization.

Transfer Learning and Multimodal Data Fusion: Transfer learning is a promising approach for improving the performance of deep learning models for autism diagnosis. By using pre-trained models, such as VGGFace or deep learning models trained on large emotion identification datasets, researchers can leverage the knowledge learned from these models to improve the performance of their own models on smaller datasets. Additionally, incorporating multiple data modalities, such as audio or textual data, can further improve the accuracy and robustness of the model by providing a more comprehensive understanding of the child's emotions and expressions.

Explainable Deep Learning and Clinical Expert Collaboration to Improve Autism Diagnosis: To improve the interpretability of deep learning models for autism diagnosis, researchers can use explainable deep learning techniques, such as attention mechanisms or XAI approaches. This will help to understand which elements of the input data are most relevant for the model's decisions. Additionally, collaboration with clinical experts is essential to collect and label data that is representative of real-world clinical assessments. These experts can provide valuable insights into which features and predictions are clinically meaningful. By focusing on improving interpretability and aligning the model with clinical assessments, researchers can develop a more reliable and trustworthy tool for autism diagnosis.

Ethical and User-Centric Considerations: Ethical concerns and user-centric considerations are essential for developing autism diagnosis systems that are responsible and trustworthy. This includes implementing strong security measures and anonymization techniques to protect sensitive data, obtaining informed consent from participants, developing bias mitigation strategies, conducting fairness and transparency audits, and designing user-friendly interfaces that comply with ethical and legal standards for medical applications.

Ensuring that the system is accessible and respectful of users' needs and concerns is fundamental to the future direction of this research.

Longitudinal Studies and Community Involvement: To obtain a thorough grasp of the model's efficacy, longitudinal studies monitoring kids' growth and development over time can be carried out. Evaluating the model's capacity to predict autism at various ages is essential to its practical usefulness. By holding workshops and conducting surveys, you can interact with communities and organizations that support people with autism to learn more about their unique needs, worries, and expectations helping in establishing trust and hence directing the course of the study. Participation in the community can yield insightful information and support the responsible and moral development of the model.

Impact Assessment and Comparison with Existing Tools: An important aspect of future research is assessing the long-term impact of using AI-based tools on the well-being and development of children. Based on model predictions, assess the effectiveness of early interventions. Additionally, conducting extensive comparative studies to determine how well the model's predictions match up with established diagnostic tools. Set benchmarks and metrics for assessing the model's performance in a clinical setting.

7.3 Outcomes:

1. Book Chapter “Advancements in AI for Mental Health: Exploring ASD, ADHD & Schizophrenia, Video Datasets and Future Directions” accepted in the book titled “Computational Intelligence for Oncology and Neurological Disorders: Current Practices and Future Directions” by Taylor and Francis.

2. Journal Publication in progress.

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